Maintenance Schedule

Diesel-electric drilling unit
EPA/CARB
TE12V4000A1N
12V4000T25L
Application group 3B

MSE50816/01E

A Rolls-Royce solution

System designation	Engine model	Hz	kW	rpm	Application group
TE12V4000A1N	12V4000T25L	60 Hz	1364 kW	1800 rpm	3B, continuous service, variable load, ICXN

Table 1: Applicability

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emissions regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Maintenance tasks that are mandatory to comply with the emission regulations are marked accordingly.

Although the warranty obligations are not dependent upon the use of any particular brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only components approved by Rolls-Royce Solutions or their equivalent. Emissions-relevant components include amongst others control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all components of the exhaust aftertreatment systems. The use of spare parts that are not of equivalent quality to components approved by Rolls-Royce Solutions may impair the effectiveness of emissions control systems. Maintenance, replacement, or repair of emission control equipment and systems may be performed by any repair facility or person of the owner's choice.

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

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Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
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Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
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1.4 Load profile

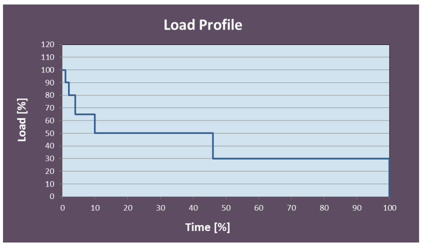
Load profile								
Load profile		Load factor %	Load indicator %	0 to 125 kW/cyl.				
Load %	Time %							
110 (100-110)	–	0 to 42	≤ 3	30,000	–	–	–	
100 (90-100)	1							
90 (80-90)	1							
80 (65-80)	2							
65 (50-65)	6							
50 (30-50)	36							
30 (5-30)	54							
Idle Speed (0-5)	–							

Table 2: Load profile

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1.5 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

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1.6 Maintenance Schedule Matrix 30000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00112	1 DAY																														
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00174	1 WK																														
WM01698	1 YR																														
WM00067	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00664	3 YR						X						X						X						X						X
WM00776	3 YR						X						X						X						X						X
WM00094	18 YR		X				X						X						X						X						X
WM00282	18 YR						X						X						X						X						X
WM00292	18 YR						X						X						X						X						X
WM00150	6 YR										X										X										X
WM00151	6 YR										X										X										X
WM00089	18 YR												X												X						
WM00090	18 YR												X												X						
WM00098	18 YR												X												X						
WM00110	18 YR												X												X						
WM00091	18 YR										X										X										X
WM00008	6 YR																														X
WM00054	18 YR																														X
WM00056	18 YR																														X
WM00062	18 YR																														X
WM00093	18 YR																														X
WM00096	18 YR																														X
WM00164	18 YR																														X
WM00921	18 YR																														X
WM00970	18 YR																														X
WM00971	18 YR																														X
WM00973	18 YR																														X
WM00974	18 YR																														X
WM00976	18 YR																														X
WM00981	18 YR																														X
WM00986	18 YR																														X
WM00006	6 YR																														

DAY = Days
WK = Weeks
YR = Years
MON = Months
HRS = Hours

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